

Placement Portal Project Report

1. Student Details

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About Me: My dream is to create applications on the web that help to sort out the everyday issues of the users while at the same time enhancing the human experience when using the technology. The most important thing when creating the application is to make sure it's based on the most up-to-date technology.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

In order to coordinate all placement activities relating to companies, students and placement drives, institutes must create an efficient system that supports each of these teams and their needs for effective communication. Currently institutes are struggling to coordinate all aspects of student placement (approvals, etc.) because they utilize multiple disparate systems, including spreadsheets, email and direct interaction. As a result, there is no efficient way to manage both how students register, and create mechanisms to prevent duplicate registrations.

This project will create a more effective solution in the form of a Placement Portal, which will allow an Admin, Companies and Students to effectively interact with one another.

Approach:

This application is made with a modular structure that relies on Flask with an RBAC (Role-Based Access Control) architecture. Some of the modules that are included include Authentication, Application Admin, and Admin of Application.

For the backend, SQLAlchemy is used as the ORM to facilitate effective communication with the database whereas Bootstrap is used at the frontend to make sure the application is responsive. It has been structured in such a way that more functionalities can be added through modular scaling (such as analytics and notifications).

3. AI/LLM Declaration

Utilization of AI technology in developing Flask applications at this time is anticipated to be approximately 15-20%. As for the remaining work, it was completed via manual means.

The following uses for ChatGPT (GPT-5) were performed:

- Identifying errors with Flask route instructions
- Forming models within a database
- Improving structure/flow of code.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
SQLAlchemy	Database ORM
SQLite	Database storage
Jinja2	Template rendering
Bootstrap	Frontend design
Flask-Login	Authentication system
HTML/CSS	UI structure
JavaScript	Client-side interactions

5. Database Schema / ER Diagram

Tables:

Tables:

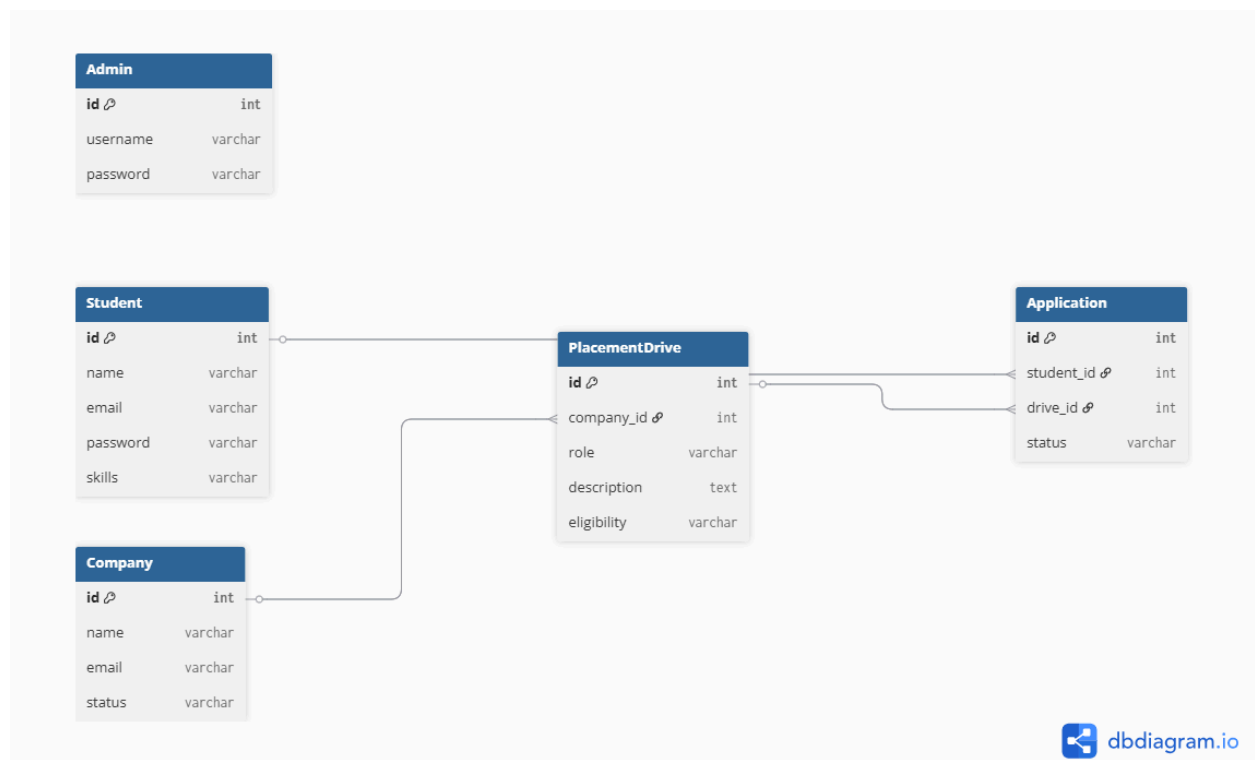
1. **Admin**
 - id, username, password
2. **Student**

- id, name, email, password, skills
- 3. Company**
 - id, name, email, status
- 4. PlacementDrive**
 - id, company_id, role, description, eligibility
- 5. Application**
 - id, student_id, drive_id, status

Relationships:

- One-to-Many → Company → Placement Drive
- One-to-Many → Student → Application
- One-to-Many → Placement Drive → Application

(Insert ER diagram image here – can be created using dbdiagram.io or draw.io or any other website)



(example taken form dbdiagram.io)

6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	Authenticate users
/register	POST	Register student/company
/admin/dashboard	GET	View system statistics
/company/add_drive	POST	Add placement drive

YAML API Definition File:

Included separately in the submission ZIP as [api.yaml](#).

7. Architecture and Features (optional)

Architecture Overview:

- [app.py](#) – Main Flask application
- [/models](#) – Database models
- [/routes](#) – Application routes
- [/templates](#) – HTML pages
- [/static](#) – CSS and JavaScript files

Implemented Features:

- User Authentication ([For All Users Including](#) Admin, Student [and](#) Company).
- [Companies Approvals](#) Process for [Registration](#).
- [Placement Drives Process](#).
- Student Application Process.
- [Statistics](#) Dashboard [for Admin](#).
- Session [Management](#).
- Responsive [UI using Bootstrap](#).

Additional Features:

- A system that monitors the user's status
- User rights should be managed correctly
- Improve the user's experience through the use of an efficient interface
- Scalable architecture on the back end

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1qgGz8-kQ_6gGSzJAoQnTm73rwNOfWwl_/view?usp=sharing
